ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CNTT & TT

BÁO CÁO BÀI TẬP
HOME WORK 1

Họ tên: Vũ Tiến Lợi

Mã số sinh viên: 20235768

Mã lớp: 166155

Hà Nội, Ngày 4 tháng 3 năm 2026

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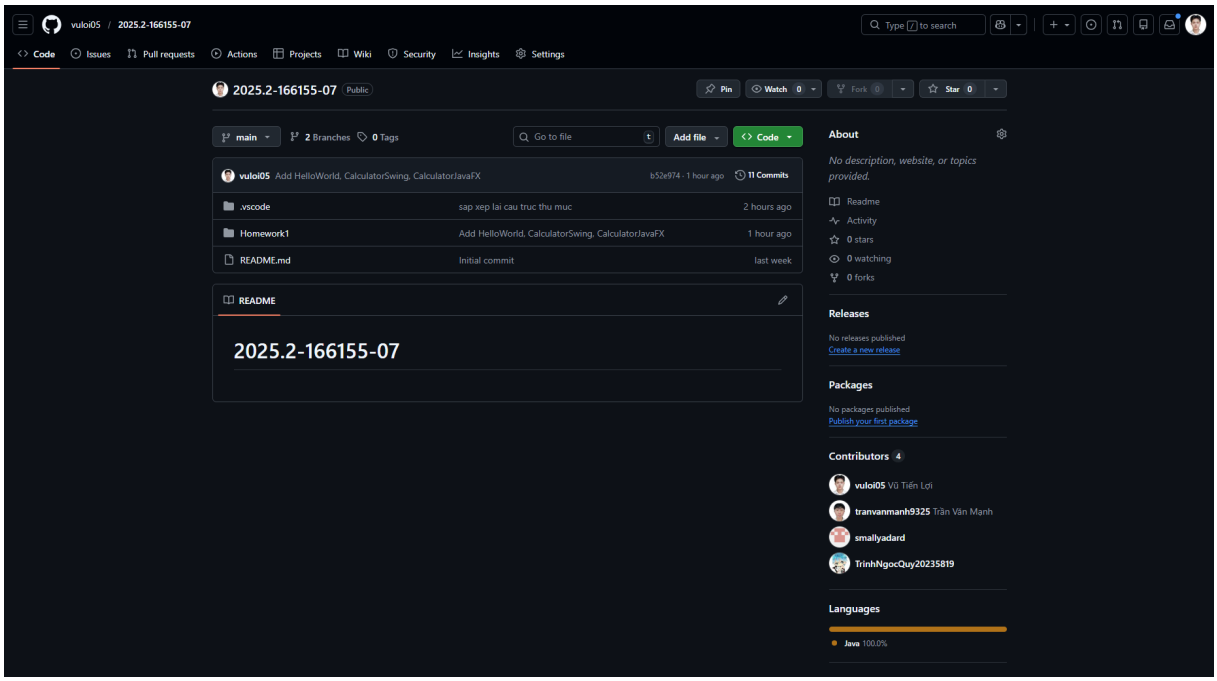
1 Giới thiệu

Bảng 1: Môi trường phát triển

Thành phần	**Version**
JDK	OpenJDK 25.0.1
JavaFX SDK	25.0.2
IDE	VS Code
OS	Windows
Git	2.52.0

2 Thiết lập GitHub Repository

Link đến repository: github.com/vuloi05/2025.2-166155-07



3 Bài tập code

3.1 HelloWorld

Chương trình hỏi tên và hiển thị lời chào

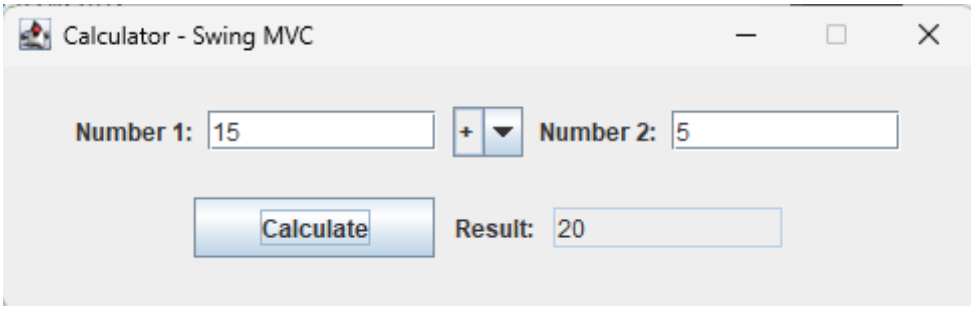
```java
import java.util.Scanner;

public class HelloWorld {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Please enter your name: ");
        String name = scanner.nextLine();

        System.out.println("Hello, " + name + "!");

        scanner.close();
    }
}
```

Kết quả:

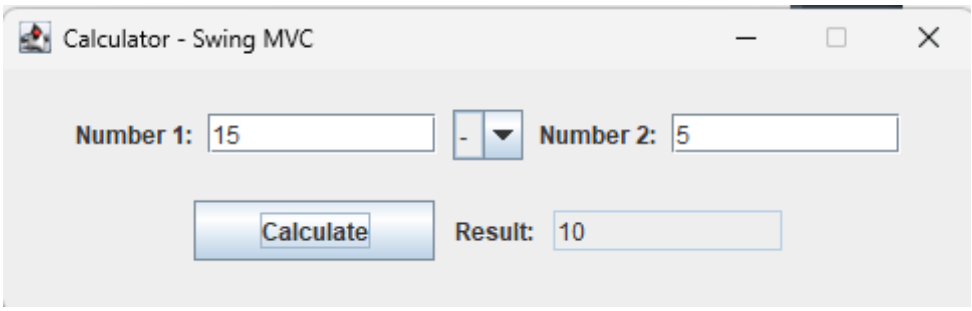
```
PS C:\DU_AN\2025.2-166155-07\Homework1\20235768_VuTienLoi>  &
-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\loivt'
dt_ws\jdt.ls-java-project\bin' 'HelloWorld'
Please enter your name: Vu Tien Loi
Hello, Vu Tien Loi!
PS C:\DU_AN\2025.2-166155-07\Homework1\20235768_VuTienLoi>
```

3.2 Calculator Swing (MVC)

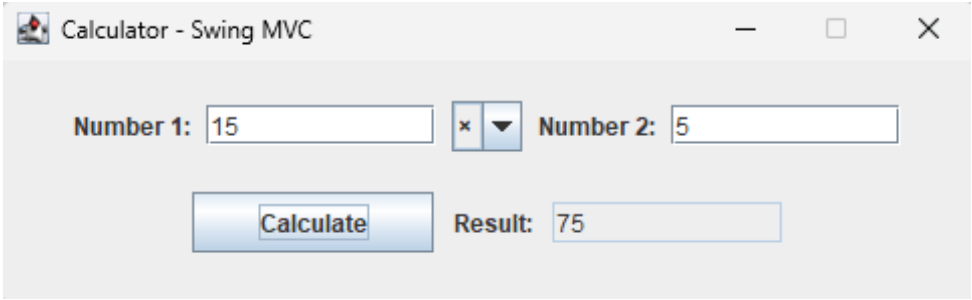
Máy tính 4 phép tính sử dụng Swing + MVC



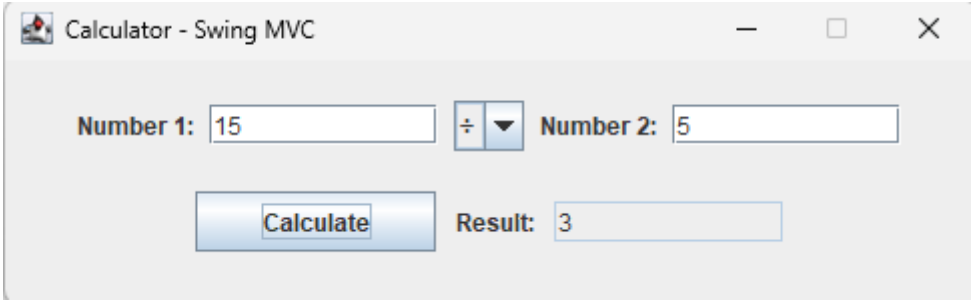
Hình 1: Phép cộng



Hình 2: Phép trừ



Hình 3: Phép nhân



Hình 4: Phép chia

3.3 Calculator JavaFX (MVC + Scene Builder)

Máy tính tương tự nhưng dùng JavaFX và FXML

4 Thao tác Git

```
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/CalculatorSwing/
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/CalculatorJavaFX/
PS C:\DU_AN\2025.2-166155-07> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Homework1/20235768_VuTienLoi/CalculatorJavaFX/bin/calculator/Main.class
        new file:   Homework1/20235768_VuTienLoi/CalculatorJavaFX/bin/calculator/view/calculator.fxml
        new file:   Homework1/20235768_VuTienLoi/CalculatorJavaFX/src/calculator/Main.java
        new file:   Homework1/20235768_VuTienLoi/CalculatorJavaFX/src/calculator/controller/CalculatorController.java
        new file:   Homework1/20235768_VuTienLoi/CalculatorJavaFX/src/calculator/model/CalculatorModel.java
        new file:   Homework1/20235768_VuTienLoi/CalculatorJavaFX/src/calculator/view/calculator.fxml
        new file:   Homework1/20235768_VuTienLoi/CalculatorSwing/bin/calculator/Main.class
        new file:   Homework1/20235768_VuTienLoi/CalculatorSwing/bin/calculator/controller/CalculatorController$CalculateListener.class
        new file:   Homework1/20235768_VuTienLoi/CalculatorSwing/bin/calculator/controller/CalculatorController.class
        new file:   Homework1/20235768_VuTienLoi/CalculatorSwing/bin/calculator/model/CalculatorModel.class
        new file:   Homework1/20235768_VuTienLoi/CalculatorSwing/bin/calculator/view/CalculatorView.class
        new file:   Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/Main.java
        new file:   Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/controller/CalculatorController.java
        new file:   Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/model/CalculatorModel.java
        new file:   Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/view/CalculatorView.java
        new file:   Homework1/20235768_VuTienLoi/HelloWorld/bin/HelloWorld.class
        new file:   Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Homework1/20235768_VuTienLoi/MVCTutorial/
```

Hình 5: Add file

```
PS C:\DU_AN\2025.2-166155-07> git add .
PS C:\DU_AN\2025.2-166155-07> git commit -m "sap xep lai cau truc thu muc"
[main 63c86ac] sap xep lai cau truc thu muc
 3 files changed, 52 insertions(+)
 create mode 100644 .vscode/launch.json
 create mode 100644 .vscode/settings.json
 rename "HW01_Tr\341\273\213nh Ng\341\273\215c Qu\303\275_20235819/HelloWorld/HelloWorld.java" => Homework1/20235819_TrinhNgocQuy/HelloWorld/HelloWorld.java (100%)
```

Hình 6: Commit

```
PS C:\DU_AN\2025.2-166155-07> git push origin main
Enumerating objects: 46, done.
Counting objects: 100% (45/45), done.
Delta compression using up to 16 threads
Compressing objects: 100% (31/31), done.
Writing objects: 100% (42/42), 11.10 KiB | 947.00 KiB/s, done.
Total 42 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/vuloi05/2025.2-166155-07.git
   9b4d88b..0db3404  main -> main
```

Hình 8: Push

```
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Update HelloWorld with welcome message"
[main 56565af] Update HelloWorld with welcome message
 1 file changed, 2 insertions(+)
PS C:\DU_AN\2025.2-166155-07> git push origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 629 bytes | 314.00 KiB/s, done.
Total 7 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/vuloi05/2025.2-166155-07.git
   b52e974..56565af  main -> main
PS C:\DU_AN\2025.2-166155-07>
```

Hình 9: Modify file

```
PS C:\DU_AN\2025.2-166155-07> git pull
remote: Enumerating objects: 66, done.
remote: Counting objects: 100% (66/66), done.
remote: Compressing objects: 100% (45/45), done.
remote: Total 64 (delta 5), reused 64 (delta 5), pack-reused 0 (from 0)
Unpacking objects: 100% (64/64), 15.44 KiB | 62.00 KiB/s, done.
From https://github.com/vuloi05/2025.2-166155-07
   63c86ac..9b4d88b  main       -> origin/main
Merge made by the 'ort' strategy.
 Homework 01_Tran_Van_Manh/.idea/workspace.xml      |  87 ++++++++++++++
 Homework 01_Tran_Van_Manh/HelloWorld.java          |  11 ++
 .../calculator_JavaFX/README.md                    | 125 +++++++++++++++++++++++
 .../controller/CalculatorController.java           |  56 +++++++++
 .../calculator_JavaFX/main.java                    |  24 ++++
 .../calculator_JavaFX/model/CalculatorModel.java   |  19 ++++
 .../calculator_JavaFX/pom.xml                      |  81 +++++++++++++
 .../main/java/controller/CalculatorController.java |  56 +++++++++
 .../calculator_JavaFX/src/main/java/main.java      |  24 ++++
 .../src/main/java/model/CalculatorModel.java       |  19 ++++
 .../src/main/resources/view/calculator.fxml        |  31 +++++
 .../classes/controller/CalculatorController.class  | Bin 0 -> 2127 bytes
 .../calculator_JavaFX/target/classes/main.class    | Bin 0 -> 1304 bytes
 .../target/classes/model/CalculatorModel.class     | Bin 0 -> 646 bytes
 .../target/classes/view/calculator.fxml            |  31 +++++
 .../compile/default-compile/createdFiles.lst       |   3 +
 .../compile/default-compile/inputFiles.lst         |   3 +
 .../calculator_JavaFX/view/calculator.fxml         |  32 ++++++
 .../calculator_swing/.vscode/launch.json           |  14 +++
 .../calculator_swing/.vscode/tasks.json            |  40 +++++++
 .../controller/CalculatorController.class          | Bin 0 -> 2085 bytes
 .../controller/CalculatorController.java           |  37 ++++++
 .../calculator_swing/main.class                    | Bin 0 -> 431 bytes
 .../calculator_swing/main.java                     |   8 ++
 .../calculator_swing/model/CalculatorModel.class   | Bin 0 -> 553 bytes
 .../calculator_swing/model/CalculatorModel.java    |  11 ++
 .../calculator_swing/view/CalculatorView.class     | Bin 0 -> 1252 bytes
 .../calculator_swing/view/CalculatorView.java      |  37 ++++++
 .../mvc/controller/HelloController.java            |  20 ++++
 Homework 01_Tran_Van_Manh/mvc/main.java            |  12 ++
 .../mvc/model/HelloModel.java                      |  13 +++
 Homework 01_Tran_Van_Manh/mvc/view/HelloView.java  |  16 +++
 32 files changed, 810 insertions(+)
```

Hình 7: Enter Caption

```
PS C:\DU_AN\2025.2-166155-07> echo "temp" > Homework1/20235768_VuTienLoi/temp.txt
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/temp.txt
PS C:\DU_AN\2025.2-166155-07> git commit -m "Add temp file"
[main 8f25291] Add temp file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Homework1/20235768_VuTienLoi/temp.txt
PS C:\DU_AN\2025.2-166155-07> git rm Homework1/20235768_VuTienLoi/temp.txt
rm 'Homework1/20235768_VuTienLoi/temp.txt'
PS C:\DU_AN\2025.2-166155-07> git commit -m "Remove temp file"
[main 0bbbc97] Remove temp file
 1 file changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 Homework1/20235768_VuTienLoi/temp.txt
PS C:\DU_AN\2025.2-166155-07> git push origin main
Enumerating objects: 41, done.
Counting objects: 100% (41/41), done.
Delta compression using up to 16 threads
Compressing objects: 100% (30/30), done.
Writing objects: 100% (35/35), 2.76 KiB | 565.00 KiB/s, done.
Total 35 (delta 16), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (16/16), completed with 3 local objects.
To https://github.com/vuloi05/2025.2-166155-07.git
   56565af..0bbbc97  main -> main
```

Hình 10: Remove file

```
PS C:\DU_AN\2025.2-166155-07> git branch loi-update
PS C:\DU_AN\2025.2-166155-07> git checkout loi-update
Switched to branch 'loi-update'
PS C:\DU_AN\2025.2-166155-07> git branch
* loi-update
  main
PS C:\DU_AN\2025.2-166155-07>
```

Hình 11: Create branch

```
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Update greeting message on feature branch"
[loi-update c5f05e8] Update greeting message on feature branch
 1 file changed, 1 insertion(+), 2 deletions(-)
PS C:\DU_AN\2025.2-166155-07> git push origin loi-update
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 639 bytes | 639.00 KiB/s, done.
Total 7 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
remote:
remote: Create a pull request for 'loi-update' on GitHub by visiting:
remote:      https://github.com/vuloi05/2025.2-166155-07/pull/new/loi-update
remote:
To https://github.com/vuloi05/2025.2-166155-07.git
 * [new branch]      loi-update -> loi-update
```

Hình 12: Thay đổi file

```
PS C:\DU_AN\2025.2-166155-07> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\DU_AN\2025.2-166155-07> git merge feature-update
merge: feature-update - not something we can merge
PS C:\DU_AN\2025.2-166155-07>
```

Hình 13: Gộp nhánh

```
HelloWorld > src > HelloWorld.java > ...
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
<<<<<<< HEAD    Syntax error on tokens, delete these tokens (Current Change)
// Main branch version
=======
// Conflict branch version
>>>>>>> conflict-branch-loi (Incoming Change)

import java.util.Scanner;

public class HelloWorld {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print(s: "Please enter your name: ");
        String name = scanner.nextLine();

        System.out.println("Hello, " + name + "!");

        System.out.println(x: "Welcome to Homework01!");

        scanner.close();
    }
}
```

Hình 14: Gộp nhánh có lỗi - 1

```
HelloWorld > src > HelloWorld.java > ...
// Main branch version    HelloWorld.java is a non-project file, only syntax errors are reported

import java.util.Scanner;

public class HelloWorld {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print(s: "Please enter your name: ");
        String name = scanner.nextLine();

        System.out.println("Hello, " + name + "!");

        System.out.println(x: "Welcome to Homework01!");

        scanner.close();
    }
}
```

Hình 15: Gộp nhánh có lỗi - 2

```
PS C:\DU_AN\2025.2-166155-07> git branch
  conflict-branch-loi
  loi-update
* main
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Add comment on main branch"
[main 787e387] Add comment on main branch
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\DU_AN\2025.2-166155-07> git checkout -b conflict-branch-loi
fatal: a branch named 'conflict-branch-loi' already exists
PS C:\DU_AN\2025.2-166155-07> git checkout conflict-branch-loi
Switched to branch 'conflict-branch-loi'
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Add comment on conflict branch"
On branch conflict-branch-loi
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Homework1/20235768_VuTienLoi/.vscode/
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/Main.class
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/controller/CalculatorController$CalculateListener.class
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/controller/CalculatorController.class
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/model/CalculatorModel.class
        Homework1/20235768_VuTienLoi/CalculatorSwing/src/calculator/view/CalculatorView.class

nothing added to commit but untracked files present (use "git add" to track)
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Add comment on conflict branch"
[conflict-branch-loi 3174364] Add comment on conflict branch
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\DU_AN\2025.2-166155-07> git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)
PS C:\DU_AN\2025.2-166155-07> git merge conflict-branch
merge: conflict-branch - not something we can merge
PS C:\DU_AN\2025.2-166155-07> git merge conflict-branch-loi
Auto-merging Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
CONFLICT (content): Merge conflict in Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
Automatic merge failed; fix conflicts and then commit the result.
PS C:\DU_AN\2025.2-166155-07> git add Homework1/20235768_VuTienLoi/HelloWorld/src/HelloWorld.java
PS C:\DU_AN\2025.2-166155-07> git commit -m "Resolve merge conflict"
[main b72bb62] Resolve merge conflict
PS C:\DU_AN\2025.2-166155-07>
```

Hình 16: Gộp nhánh có lỗi - 3